Source-C architecture and proxy exposure

summary of the 2026-10-21 to 2026-10-26 investigation

local_mixing project notes

Prepared 2026-10-26; campaign still running (Section 9)

1 Questions

RC asked three things after the C-restructuring experiments (`C_RESTRUCTURING_EXPERIMENTS_20261021.md`):

1. Why does the balanced C (every internal value read exactly twice as a control, write counts balanced at every cut) leak as many internal values after stage 6 as the random C, and is there any parameter on which it is better?
2. Is there a way to design C so that fewer of its internal values are exposed, without growing it or losing the pseudorandomness of the function it computes?
3. Test with the latest stage-2 compute (three-mask, modified) and the latest stage 6 (adaptive, v2), on the server.

Short answers: (1) exposure is a per-control-read phenomenon, the balanced C only redistributes the 2,000 reads, and it does win on read-side metrics (gates with both controls exposed are halved) while losing on the count; (2) within g57 circuits of 1,000 gates on 64 wires, no: every read distribution that lowers the count breaks avalanche, because the random C already sits at the minimum number of read values that full diffusion needs; (3) under three-mask + v2 every C gives 0/0 in the census, and the C-blind windowed observers on the stage-6 outputs reproduce exactly the same per-read law, so C design redistributes exposure there as well.

2 What was measured, and where in the pipeline

Pipeline per arm: stage 1 sandwich (C on 64 wires, 1,000 g57 gates, plus D and slicing, 2,832 gates) → stage 2 compute block (either the pinned baseline `managed_tdp`, ∼778k gates, or the three-mask compute with RC's modifications `--random-helpers --unique-cover`, ∼505k gates) → stages 3–5 (DB mixing, split, crossing, reduced profile 2,2,2,2,2; ∼5.5M or ∼3.5M gates) → stage 6 adaptive resynthesis, policy *asis* (one seed, keep-smaller) or *v2* (two seeds, keep-smaller-and-not-leakier, moved boundaries, size-scaled budgets). One master seed is shared by every arm, so D, the sandwich and all stage seeds are identical; only C (and the stage-2 block) differ.

Measurement families, all on the honest-slice inputs and all re-validated on 8,192 fresh plus structured inputs:

- **Census** (C-aware, every cut, every wire): a C birth is *directly* exposed if some wire version equals it or its complement; *affinely* exposed if it is an affine combination of simultaneously live wires. Reported on stage-6 outputs; every stage-5 circuit in this study has 0/0.
- **Narrow portfolio** (C-blind): nine windowed resynthesis attacks on a circuit (raw, factor, factor_perm, factor_shift63, factor_w250, pairs, payload factor_w125 / factor_w250 / delta_w250), then an observer scores the attacked circuit; counts are deeper C values found at ≥ 2 locations (R) or once (O). "Deeper" excludes the public-quadratic boundary births (60 for the original C, 49 balanced, 73 burst).

- **C-blind adaptive resynthesis** (resilience step): stage 6 with no C information (largest feasible blocks, keep-smaller) run to plateau on an output, census after each round.

Three C's: the campaign's random C (geometric read distribution: 336 of 936 internal births never read as a control, 217 read once, 130 twice, 253 hubs with ≥ 3 reads), RC's balanced C (reproduced byte-exact, sha `d0586653...`), and a sticky-target "burst" C ($q = 0.67$: 738 of 937 births unread, 104 hubs with 8+ reads) built as a model test; its avalanche is broken on purpose, so it is a diagnostic, not a candidate.

3 Exposure is per control read; unread values are immune

Validated affine exposure of the internal births after one stage-6 round, by source control fanout (number of gates that read the value before it is overwritten):

run (baseline stage 2)	fanout 0	1	2	3	4	5+	total
RC original C, adaptive pass 1	7/336 (2%)	65/217 (30%)	52/130 (40%)	45/71 (63%)	43/59 (73%)	107/123 (87%)	319
our original C, asis round 1	4/329 (1%)	59/201 (29%)	64/150 (43%)	54/88 (61%)	36/57 (63%)	95/111 (86%)	312
our original C, asis round 4	12/329 (4%)	87/201 (43%)	96/150 (64%)	69/88 (78%)	53/57 (93%)	106/111 (95%)	423
RC balanced C, adaptive pass 1	–	–	455/936 (49%)	–	–	–	455

Direct exposure has the same shape at lower levels (1 / 21 / 28 / 42–46 / 47–51 / 62–88 %). Each read is roughly an independent chance of about 0.3 (affine) or 0.2 (direct) that the synthesizer materialises the value: it only needs a C value in the clear when it re-implements a gate that multiplies it, whereas a value that is only ever a target accumulates inside its masked wire. Lifetime adds nothing once fanout is fixed (RC's 52-feature analysis of the balanced C is at chance; in the original fanout-2 cohort the short-lived values are, if anything, more exposed). The few values exposed in the three-mask arms (3–16 after several asis rounds) are all read values.

4 Why the balanced C leaks as much, and what it improves

Reads per gate are fixed at two, so a design can only redistribute 2,000 reads over 936 internal births (plus 64 inputs). The balanced C removes the 253 hubs but also the 553 values read zero or one time, moving everything to the two-read rate. Same census, same driver (RC's adaptive pass 1), affine with direct in parentheses:

metric	original C	balanced C
exposed births	319 (239)	455 (283)
fraction of control reads whose source value is exposed	0.69 (0.55)	0.49 (0.30)
C gates with both controls exposed (inputs public; floor ∼29)	462 (307)	248 (127)
C gates whose increment is exposed (target before and after)	120 (67)	241 (96)

The count $f(k)$ is concave in the reads k, so the equal distribution maximises it; $k\,f(k)$ is convex, so the same distribution minimises read-weighted exposure and, approximately, the number of gates whose two multiplicands are both in the clear. Those two objectives are duals; that is the whole story of "balanced = same leak". It also explains the AUC 0.5: with every value at the same read count, which ones are exposed is decided downstream by the synthesizer, not by any source feature.

5 The design space: concentrating reads lowers the count but breaks avalanche

Candidate designs (seed 7) with the count predicted from the fanout rates above, and pseudorandomness proxies measured on the circuits (SAC = per input/output bit flip probability over 4,096 inputs; full-degree fraction = share of (random 12-variable subcube, output) pairs whose restriction has degree 12, 0.5 for a random function):

design	unread	pred. affine count	reads-weighted	gates both	increments	SAC max bias	full-degree
random (RC's C)	336	319	0.69	469	125	0.027	0.49
balanced (RC's C)	0	374 (obs. 455)	0.40	185 (obs. 248)	165 (obs. 241)	0.033	0.50
burst $q = 0.5$	602	225	0.81	572	68	0.36 (dead pairs)	0.50
burst $q = 0.67$	738	160	0.88	650	35	0.50	0.48
phased hot set 16/100	809	140	0.97	842	22	0.50	0.35
preferential reads $w = 4$	730	129	0.89	699	30	0.50	0.05

A random 64-wire g57 circuit needs about 1,000 gates, equivalently about 590 read values, before every input bit reaches every output bit (650 gates: dead input/output pairs; 800 gates: SAC bias 0.12; 1,000: 0.03). The random 1,000-gate C sits at that floor. Every design that lowers the exposed count does so by leaving values unread, and every such design that lowers it by more than a few percent loses avalanche. Adding gates to compensate adds births one for one. Within this family, C design redistributes exposure; it does not reduce it.

6 Server campaign: matched arms with three C's

Tree `~/cdesign` on machine1, same SEED as the matched three-mask campaign, generator extended with `--source-file`. Stage-2 sizes are unchanged by the C (777–780k baseline, 505k three-mask).

6.1 Causal test under the baseline stage 2 (asis policy)

C	round	gates in → out	direct / affine	per-fanout affine (0, 1, 2, 3, 4, 5+)	reads-w.	gates both	incr.
original	1	5,508,346 → 2,831,405	222 / 312	4/329, 59/201, 64/150, 54/88, 36/57, 95/111	0.65	412	110
original	2	2,831,405 → 1,905,790	327 / 382	9, 80, 78, 67, 43, 105	0.76	549	157
balanced	1	5,488,705 → 2,830,920	328 / 453	453/936 at fanout 2	0.48	230	236
balanced	2	2,830,920 → 1,919,514	464 / 582	582/936	0.62	381	378
burst	1	5,506,304 → 2,991,300	82 / 124	5/738, 3/30, 10/21, 12/24, 9/20, 85/104	0.79	560	26
burst	2	2,991,300 → 1,857,096	145 / 165	5/738, 10, 11, 20, 17, 102/104	0.94	712	38

Predictions from Section 5 were 374–455 exposed / 185–250 gates-both for balanced (observed 453 / 230) and about 160 / 650 / 35 for burst with the unread values absent (observed 124 / 560 / 26, with 5 of 738 unread values exposed). Same pipeline, seed, D and sandwich; only C changed, so the read structure is the cause.

6.2 Three-mask modified stage 2 with v2 stage 6

Census on every round's output: 0/0 deeper for all three C's (original: 9 rounds to plateau at 2,706,063 gates; balanced: 37 rounds so far; burst: 38 rounds; both still shrinking 0.1% per round, the two-zero-gain plateau not yet reached). Stage-5 sizes differ by at most 3% across the C's.

Narrow portfolio on the stage-5 outputs (nine cases): original 7/3 deeper R/O, balanced 14/2, burst 6/2 under three-mask; 0/0, 1/0, 2/0 under baseline stage 2. The balanced C hands the windowed observer the most repeated values because every value is read twice, so every recovered fire control has a second fire to be matched against.

6.3 The decisive comparison: C-blind narrow portfolio on the stage-6 outputs

Nine cases on outputs of equal size (original plateau output; balanced round 14; burst round 15), all 0/0 under the census:

C	gates	deeper found (R / O)	by fanout 0 / 1 / 2 / 3 / 4 / 5+	reads-w.	gates both known	incr. known
original	2,706,063	233 (157 / 76)	7 / 36 / 45 / 43 / 26 / 76	0.51	338	119
balanced	2,703,707	265 (172 / 93)	all 265 at fanout 2	0.28	151	129
burst	2,628,101	108 (76 / 32)	1 / 5 / 5 / 11 / 9 / 77	0.73	584	84
original, baseline st. 2, asis plateau	666,745	450 (444 / 6)	23 / 96 / 103 / ...	0.82	721	292

Find rates by fanout on the original C are 2 / 18 / 30 / 49 / 46 / 68 %. The balanced C's single cohort is found at 30%, the original's fanout-2 rate; the burst C's hubs at 74% with 1 of 738 unread values found. The per-read law therefore holds for C-blind observers on the compressed output exactly as it does for the C-aware census under the baseline encoding, and with the same trade-off: balanced +14% count and half the gates with both controls known; burst −54% count and 1.7× the gates with both controls known.

C-blind adaptive resynthesis of the round-21 outputs, deeper direct/affine per round (stopped after twelve rounds): balanced 1, 4, 9, 10, 10, 10, 12, 12, 13, 14, 16, 17; burst 1, 1, 0, 1, 3, 2, 2, 2, 2, 2, 2, 3; original reference 3, 4, 6, 6, 6, 6 with plateau at round 6.

6.4 Where the values appear

Direct copies in the stage-6 outputs (baseline stage 2, asis round 1): original 222 exposed values, 85 at two or more locations, 365 physical copies carrying 2,317 control reads; balanced 328 / 75 / 468 / 2,714; burst 82 / 38 / 162 / 946. No value is ever live at two locations at the same time in any run: copies are sequential re-materialisations in different blocks. Copies per value grow with fanout (original round 4: 5+ values 3.9 copies, 80% multi-location).

Fire fragments: a fire is 170 stage-2 gates, about 1,200 stage-5 gates after the 7.1× expansion, five 250-gate windows. The 20 product units are emitted in groups sharing the left factor (17–35 stage-2 gates, 120–250 at stage 5), and any group sums to $L \cdot x_v$, so one group in a window already yields x_v. That is why the 250-gate factoring finds 4–13 repeated values on stage-5 circuits while the 125-gate window finds about one, and why the finds are spread along a value's lifetime rather than clustered at its consumer fires.

7 Side findings on stages 2 and 6 (original C, matched campaign)

stage 2 / stage 6	plateau output (rounds)	census	narrow on output (6 cases)	C-blind resynth., last	raw ledger after
baseline / asis	666,745 (24)	384/424	436 / 7	444/484	410 / 59
three-mask as-is / asis	2,611,000 (16)	21/21	198 / 52	39/39	13 / 29
three-mask as-is / v2	2,368,032 (14)	0/0	229 / 46	9–11	2 / 7
three-mask modified / asis	2,761,726 (12)	14/14	104 / 53	17/17 (plateau)	2 / 15
three-mask modified / v2	2,706,063 (9)	0/0	117 / 57 (157 / 76 on 9 cases)	6/6 (plateau)	1 / 5

- Stage-6 compression exposes the mixed output to the windowed observers: 157/76 repeated/once on the v2 plateau output against 10 (7 repeated) on the stage-5 circuit it came from. The v2 acceptance rule only sees the census family. Remixing the output (stage 3 again) brings the raw ledger back to 0/0.
- v2 is no better than asis against the C-blind windowed observers; it is better in the census family (0 vs 14–21) and under C-blind adaptive resynthesis (6 vs 17–39).
- RC's two stage-2 modifications (random dirty helpers, unique covers) roughly halve the repeated count of the as-is three-mask compute on the plateau outputs (104–117 vs 198–229).

8 Conclusions

1. Exposure after resynthesis is driven by control reads of C's values: about 0.3 per read for the affine family under the baseline encoding, about 0.25 per read for the C-blind windowed observers on three-mask + v2 outputs, and essentially zero for values that are never read.
2. The balanced C does not reduce exposure because it cannot: with two reads per gate the design only chooses who absorbs the reads. It maximises the number of exposed values (+40–45%) and minimises the read-side quantities (gates with both multiplicands known halved, no exposed value serving more than two gates); which of these matters depends on what the attacker needs to reconstruct C.
3. No C design in this family lowers exposure without losing avalanche: the random 1,000-gate C already has the fewest read values compatible with full diffusion on 64 wires. Concentrating reads (bursts, hot sets, preferential reads) cuts the count by up to 60% in the model and in the measurement, and breaks avalanche in every variant tried.
4. The lever that reduces exposure is the encoding and the synthesizer policy, not C: three-mask stage 2 takes the census from 205–312 exposed values per asis round to 2–3, and v2 acceptance to 0. Against C-blind windowed observers the compressed outputs remain exposed (100–230 values) for every C and both policies; that is the open problem, and it belongs to stages 2 and 6.
5. Recommendation: keep C random; spend the effort on the stage-6 side (acceptance rules that also measure the windowed-factoring family, or a remix after the last resynthesis, which restores 0/0 on the raw ledger), and on the stage-2 fire structure (interleaving fires so that no factor group fits a 250-gate window).

9 Status, artifacts, limitations

Running on machine1 (`~/cdesign`): the balanced and burst v2 runs (rounds 37–38, plateau rule not met); their C-blind adaptive resynthesis attacks on the round-21 outputs were stopped after 12 rounds, and the other session's work in `~/threemask_cmp`. One incident: the disk filled on 2026-10-22 (narrow-portfolio observer scratch, ∼50 GB) and killed every run; all were resumed from their markers, a cleanup loop added, attack workers reduced to two.

Artifacts:

- lab note `docs/notes/c_design_exposure_20261021.md` (sections 1–9 with every table and update); refreshed `docs/notes/stage2_stage6_campaign_20261021.md`;
- server scripts under `~/cdesign/py` and `~/cdesign/py6`: `pipeline.py`, `census_by_fanout.py`, `locations.py`, `fire_windows.py`, `narrow_fanout.py`, `narrow_gates.py`, `resil_summary.py`, `early_narrow.py`;
- all scripts and the two source circuits are copied under `docs/notes/c_design_report_20261026/scripts/` (`local/expo_model.py`, `local/design_sim.py`, `server/...`); memory notes `cdesign-campaign-20261021` and `server-disk-full-observers-20261022`.

Limitations: one C per design and one master seed, so run-to-run variation is not measured (RC's and our original-C rates differ by a few points per fanout class); the burst C is a diagnostic with broken avalanche; the balanced and burst stage-6 comparisons are at equal size rather than at the plateau rule; all counts are validated candidate equalities on finite samples, not Boolean proofs; the census and narrow families measure different things and should not be added.